

Question 1.

a. Simplify

i. $\sqrt{27} + \sqrt{75}$ 1

ii. $(3\sqrt{2} + \sqrt{5})(3\sqrt{2} - \sqrt{5})$ 1

b. Write $0.\dot{2}\dot{9}$ as a fraction. 1c. Solve $2^x = 25$, writing your answer to 2 decimal places. 1d. Solve $-3 \leq 2 - 3x \leq 8$ 2e. Given $p^2 = q^2 + r^2 - 2qr \cos P$ make $\cos P$ the subject 2f. Simplify $(\sin x + \cos x)^2 - 2 \sin x \cos x$ 2g. Sketch the region where $y \leq 4 - x^2$ and $y \geq 2x$ 2

Question 2

- a. Express in simplest form 2

$$(a^2 + 8a + 15) \times \frac{a}{a+3}$$

- b. Solve the simultaneous equations 2

$$x^2 + y^2 = 25$$

$$y = x$$

- c. On separate number planes draw neat sketches of:

i. $x = y^2 - 1$ 1

ii. $y = x^3$ 1

iii. $xy = 2$ 1

- d. $Q(x)$ is the function $x^2 - 6x + 8$

i. Find the minimum value of $Q(x)$ 2

ii. Solve $Q(x) > 0$ 1

- e. Factorise and simplify $2^{n+1} + 2^n$ 2

Question 3

a. Solve

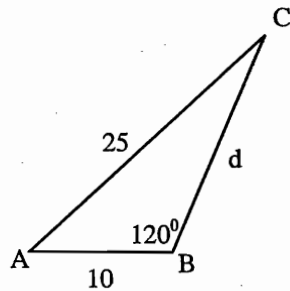
i. $\sin x = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$ 2

ii. $\tan^2 x = 1$ for all real x 2

b. If $\sin A = \frac{5}{13}$ and A lies between 90° and 360° find the exact value(s) of $\cot A$ 2

c. Sketch $y = 2\sin 3x$ for $0^\circ \leq x \leq 360^\circ$ 3

d. Find d



3

Question 4

- a. For the series $0.5 + 0.75 + 1 + \dots$

Find the sum of the first twelve terms

2

- b. Find the limiting sum of the series

$$12 + 4 + 1\frac{1}{3} + \dots$$

2

- c. The first term of an arithmetic series is 31 and the sixth term is 96.

Find the twentieth term.

2

- d. If $S_n = 11n - n^2$ find T_n

2

- e. If $x + 1$, $8 - x$, $x + 10$ are in geometric progression

Find the progression

2

- f. Evaluate $\sum_{n=1}^{12} 5 \times 2^{n-1}$

2

Question 5

a. Evaluate

i. $\lim_{x \rightarrow 0} \frac{x^2 + x}{2x}$ 1

ii. $\lim_{x \rightarrow 0} \frac{x+3}{x-3}$ 1

iii. $\lim_{x \rightarrow \infty} \frac{x+1}{2x-1}$ 1

b. What is the difference between the graphs of

$y = x + 2$ and $y = \frac{(x-2)(x+2)}{x-2}$ 2

c. Find $\frac{dy}{dx}$ given

i. $y = 2x^3 - 7x^2 + 8x - 1$ 1

ii. $y = x^2(x+3)$ 2

iii. $y = \frac{2x^2 - 3}{7 - x}$ 2

iv. $y = (5x^4 - 7)^4$ 2

Question 6

a. For the curve

$$y = 2x^3 - 3x^2 - 12x + 2$$

- i. Find the stationary points and determine their nature 3
- ii. Find the point(s) of inflexion 2
- iii. Sketch the curve showing all important features 2

b. If $f(x) = 2x^3 - 9x^2 - 60x$

Solve $f'(x) = 0$ 2

c. Find the equation of the tangent to 3

$$y = \sqrt{3x-2} \text{ at } (2,2)$$

Question 7.

- a. Calculate the acute angle between the lines 2

$$y = 2x - 3 \text{ and}$$
$$3x + 5y - 1 = 0 \text{ to the nearest minute}$$

- b. Divide A(1,7) and B(5,-2) externally in the ratio 3:2 2

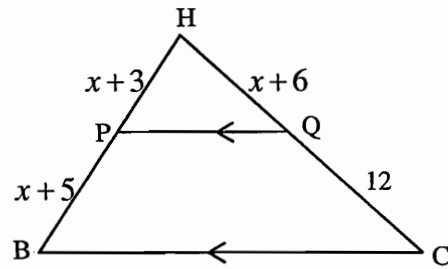
- c. Solve for x : $|2x - 5| + 3x = 0$ 3

- d. Expand $\sin(x + y)$. Hence write down an expression for $\sin 2\theta$ 2

- e. Solve the inequality $\frac{3}{x(x-2)} < 1$ 3

Question 8

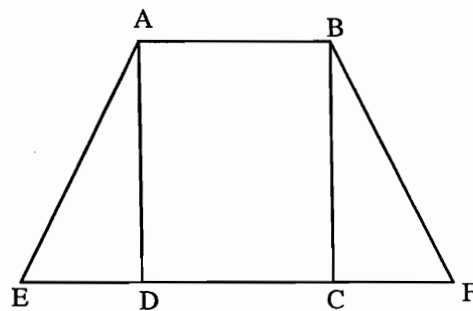
a.



3

$PQ \parallel BC$. Show x has 2 values and check these.

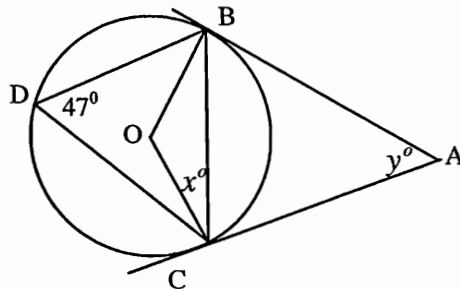
b.



3

ABCD is a rectangle. $ED = CF$
Prove $AE = BF$

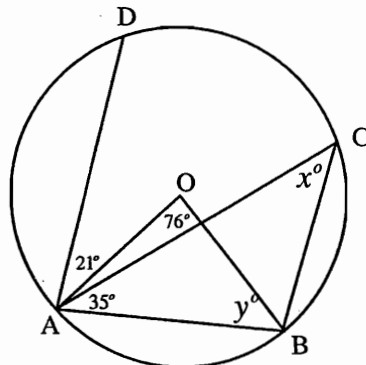
c.



3

AB, AC are tangents. Find $\angle OCB$, $\angle BAC$ giving reasons

d.



3

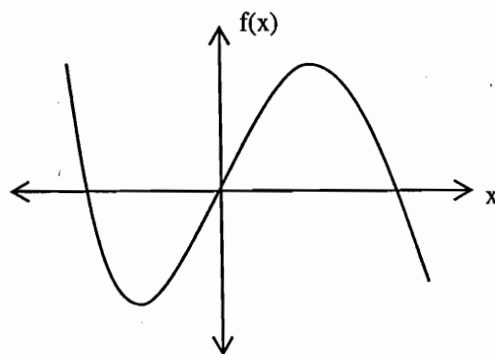
Given: $\angle DAO = 21^\circ$, $\angle AOB = 76^\circ$, $\angle ACB = x$, $\angle CAB = 35^\circ$ and $\angle OBA = y$

Find x, y and prove $AD \parallel BC$

Question 9

- a. Copy the graph of the function below onto your own paper

Directly below the graph, draw the derivative of the function, lining up important features



2

- b. Find the exact value of $\cos 75^\circ$

2

- c. Find the equation of the line that passes through the point of intersection of the lines $3x + 2y - 6 = 0$ and $x - 2y + 5 = 0$, and the point $(-3, -2)$

3

- d. From an observation tower PQ of height 50m, two points X and Y at ground level have bearings $042^\circ T$ and $130^\circ T$. From the top of the tower the angles of depression of X and Y are 32° and 48° respectively.

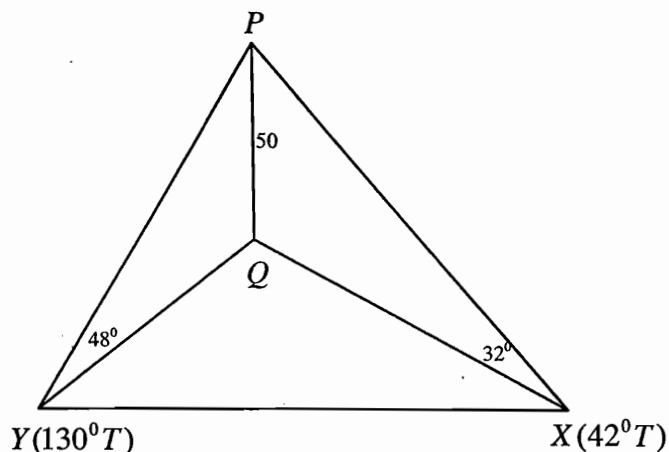
Prove that:

i. $XY^2 = 50^2 (\cot^2 32^\circ + \cot^2 48^\circ - 2 \cot 32^\circ \cot 48^\circ \cos 88^\circ)$

3

- ii. Find XY to the nearest metre.

2



Year 11 2007 Yearly Ext 1.

Q1) a) $2^x = 25$
 $x = 4.64 \text{ to } 2 \text{ DP}$

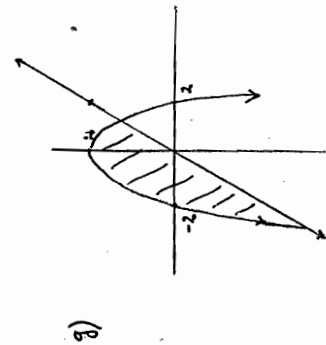
b) $\frac{29}{99}$

a) $\sqrt{3\sqrt{3} + 5\sqrt{3}}$
 $= 8\sqrt{3}$
 ii) $\frac{18-5}{13}$

d) $-3 \leq 2-3x \leq 8$
 $-5 \leq -3x \leq 6$
 $\frac{5}{3} \geq x \geq -2$

e) $\cos P = \frac{q^2 + r^2 - p^2}{2qr}$

f) $= \sin^2 x + 2 \sin x \cos x + \cos^2 x$
 $= 2 \sin x \cos x$
 $= 1$



graph
region

g)

d) $x^2 - 6x + 8 = (x-3)^2 - 1$
 $= (x-4)(x-2)$
 i) min value -1
 ii) $Q(x) > 0$
 when $x < 2$ or $x > 4$

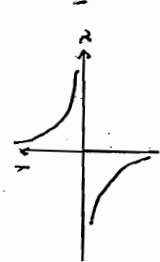
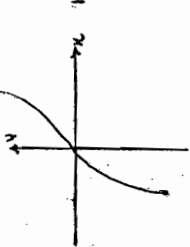
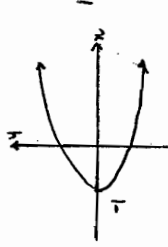
e) $2^n(2+1) = 3 \cdot 2^n$

Q2) a) $(a+s)(a+s) \times \frac{a}{a+s}$

b) $a(\arcs)$

b) $2x^2 = 25$
 $x = \pm \frac{5}{\sqrt{2}}$

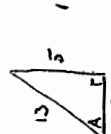
c) $\left(\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}}\right)$ $\left(-\frac{5}{\sqrt{2}}, -\frac{5}{\sqrt{2}}\right)$



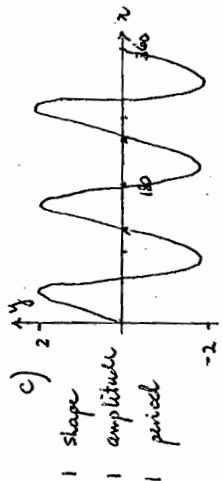
Q3) a) $\sin x = \frac{1}{2}$

$\therefore x = 30^\circ, 150^\circ$

ii) $\tan x = \pm 1$
 $\therefore x = 45^\circ, 135^\circ, 225^\circ, 315^\circ$



$\therefore \cot A = -\frac{12}{5}$



d) $\cos 120^\circ = \frac{100 + d^2 - 625}{200d}$

$10d = d^2 - 525$

$d^2 - 10d - 525 = 0$
 $d = \frac{10 \pm \sqrt{100 + 2100}}{2}$

$= 5 \pm \sqrt{550}$
 $= -5 \pm 5\sqrt{22}$

$d > 0$

$\therefore d = 18.45$

f) $5 + 5.2 + 5.4$

$S_{12} = 5(2^{12} - 1)$

$= 20475$

Q4) a) $S_2 = \frac{1}{2} \{1 + 11 \cdot \frac{1}{4}\}$
 $= \frac{1}{2} \times \frac{15}{4}$
 $= \frac{15}{8}$

b) $S_{\infty} = \frac{a}{1-r}$
 $= \frac{12}{1-\frac{1}{3}}$
 $= 12 \times \frac{3}{2}$
 $= 18$

c) $a + 5d = 96$
 $a = 31$
 $\therefore 5d = 65$
 $d = 13$

$T_{20} = 31 + 19 \cdot 13$
 $= 278$

d) $S_n = 11n - n^2$
 $S_{n-1} = 11(n-1) - (n-1)^2$
 $= 11n - 11 - n^2 + 2n - 1$
 $= 13n - n^2 - 12$
 $\therefore T_n = 11n - n^2 - 13n + 12$
 $= 12 - 2n$

e) $\frac{8-x}{x+1} = \frac{x+10}{8-x}$
 $64 - 16x + x^2 = x^2 + 11x$
 $27x = 54$
 $x = 2$

$\therefore 3, 6, 12, 24, \dots$

Q5) a) i/ $\lim_{x \rightarrow 0} \frac{x(x+1)}{2x}$

= $\frac{1}{2}$

ii/ -1

iii/ $\frac{1}{2}$

b) straight line v's straight line
with a hole at (2, 4)

c) i/ $\frac{dy}{dx} = 6x^2 - 14x + 8$

ii/ $\frac{dy}{dx} = x^2 \cdot 1 + 2x(x+3)$
= $3x^2 + 6x$

iii/ $\frac{dy}{dx} = \frac{(7-x)4x + (2x^2-3)}{(7-x)^2}$
= $\frac{28x - 4x^2 + 2x^2 - 3}{(7-x)^2}$
= $\frac{28x - 2x^2 - 3}{(7-x)^2}$

iv/ $\frac{dy}{dx} = 4(5x^4 - 7)^3 \cdot 20x^3$
= $80x^3(5x^4 - 7)^3$

Q6) a) $y = 2x^3 - 3x^2 - 12x + 2$

i/ $\frac{dy}{dx} = 6x^2 - 6x - 12$

Want $\frac{dy}{dx} = 0$ $6(x^2 - x - 2) = 0$
 $\therefore x = -1 \text{ or } +2$

$\frac{d^2y}{dx^2} = 12x - 6$

at $x = 2$ $\frac{d^2y}{dx^2} > 0 \therefore$ min at
(2, -18)

at $x = -1$ $\frac{d^2y}{dx^2} < 0 \therefore$ max at
(-1, 9)

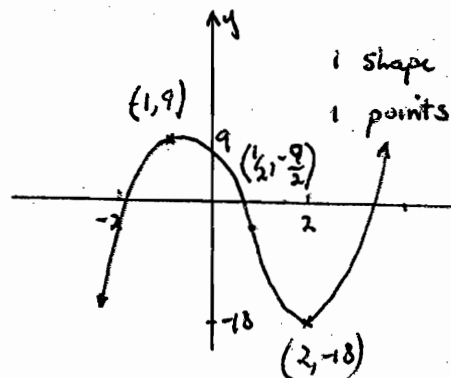
ii/ poss infl at $\frac{d^2y}{dx^2} = 0$

ie $x = \frac{1}{2}$

x 0 $\frac{1}{2}$ 1

$\frac{d^2y}{dx^2}$ < 0 0 > 1

$\therefore$ infl at $(\frac{1}{2}, \frac{9}{2})$



b) $f'(x) = 6x^2 - 18x - 60$
= $6(x^2 - 3x - 10)$
= $6(x-5)(x+2)$

$f'(x) = 0$ when $x = -2$ or 5

Q6C)

$y = (3x-2)^{1/2}$

$\frac{dy}{dx} = \frac{1}{2}(3x-2)^{-1/2} \cdot 3$

at $x = 2$ $\frac{dy}{dx} = \frac{3}{2} \times \frac{1}{2} = \frac{3}{4}$

$y - 2 = \frac{3}{4}(x-2)$

$4y = 3x + 2$

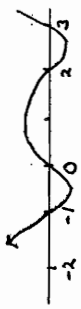
Q7)

a) $m_1 = 2$ $m_2 = -\frac{3}{5}$

$\tan(\alpha - \beta) = \left| \frac{2 + \frac{3}{5}}{1 - 2 \cdot \frac{3}{5}} \right|$
 $= \left| \frac{\frac{13}{5}}{\frac{1}{5}} \right|$

$\alpha - \beta = 85^\circ 37'$

b) $3x(x-2) < x^2(x-2)^2$
 $0 < x(x-2)(x^2-2x-3)$
 $0 < x(x-2)(x-3)(x+1)$



$\therefore x < -1$ or $0 < x < 2$ or $x > 3$

c) $|2x-5| = -3x$

$2x-5 = 3x$ $-2x+5 = -3x$
 $5x = +5$ $x = -5$
 $x = +1$

Do not work

d) $\sin(x+y) = \sin x \cos y + \sin y \cos x$
 $\sin 20 = 2 \sin \theta \cos \theta$

b) A (1,7) B(5,-2) 3:-2
 $\left(\frac{1-2}{1}, \frac{7-2}{1} \right)$
 $(13, -20)$

Note: No reasons needed for x or y.

Q8) a)

$\frac{x+3}{x+5} = \frac{x+6}{12}$

$12x+36 = x^2+11x+30$

$0 = x^2+x-6$

$= (x-3)(x+2)$

$\therefore x = 3$ or -2

both work

b)

In $\triangle ADE \sim \triangle BCF$

$ED = CF$ given

$AD = BC$ prop of rect.

$\angle EDA = \angle FCB = 90^\circ$

as $\angle ADE = \angle BCD = 90^\circ$ prop of rect.

$\therefore \triangle ADE \cong \triangle BCF$

$\therefore AE = BF$ corresp sides

of $\triangle$ s equal

c)

$\angle COB = 94^\circ$ (angle at centre)

$\angle COB = 43^\circ$ (angle at centre)

$\therefore \angle CAB = 86^\circ$

$\angle OBA = 90^\circ = \angle OCA$

d)

$\angle = 38^\circ$ (angle at centre $2x$ angle at circumference)

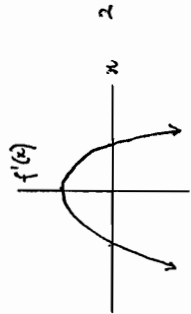
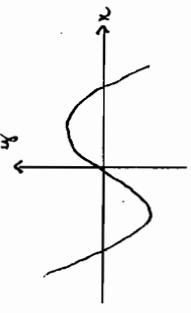
$\angle = 52^\circ$ ($\triangle OAB$ isos)

$\angle AOC$ isos $\therefore \angle OAC = 14^\circ$

$\angle DAB + \angle ABC = 180$

$\therefore AD \parallel BC$

Q9) a)



b) $\cos(45+30) = \cos 45 \cos 30 - \sin 45 \sin 30$
 $= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$

$= \frac{\sqrt{3}-1}{2\sqrt{2}}$

c) $(3x+2y-6) + \lambda(x-2y+5) = 0$
 $-9-4-C + \lambda(-3+4+5) = 0$

$6\lambda = 19$

$\lambda = \frac{19}{6}$

$6(3x+2y-6) + 19(x-2y+5) = 0$

$18x+12y-36+19x-38y+95 = 0$

$37x-26y+59 = 0$

d) $\frac{50}{yQ} = \tan 48^\circ \therefore yQ = 50 \cot 48^\circ$

$\frac{50}{xQ} = \tan 32^\circ \therefore xQ = 50 \cot 32^\circ$

$\therefore \triangle QYX$

$XY^2 = 50^2(\cot^2 48^\circ + \cot^2 32^\circ) = 50^2 \cot 88^\circ$

ii) $XY = 50 \cot 44^\circ$